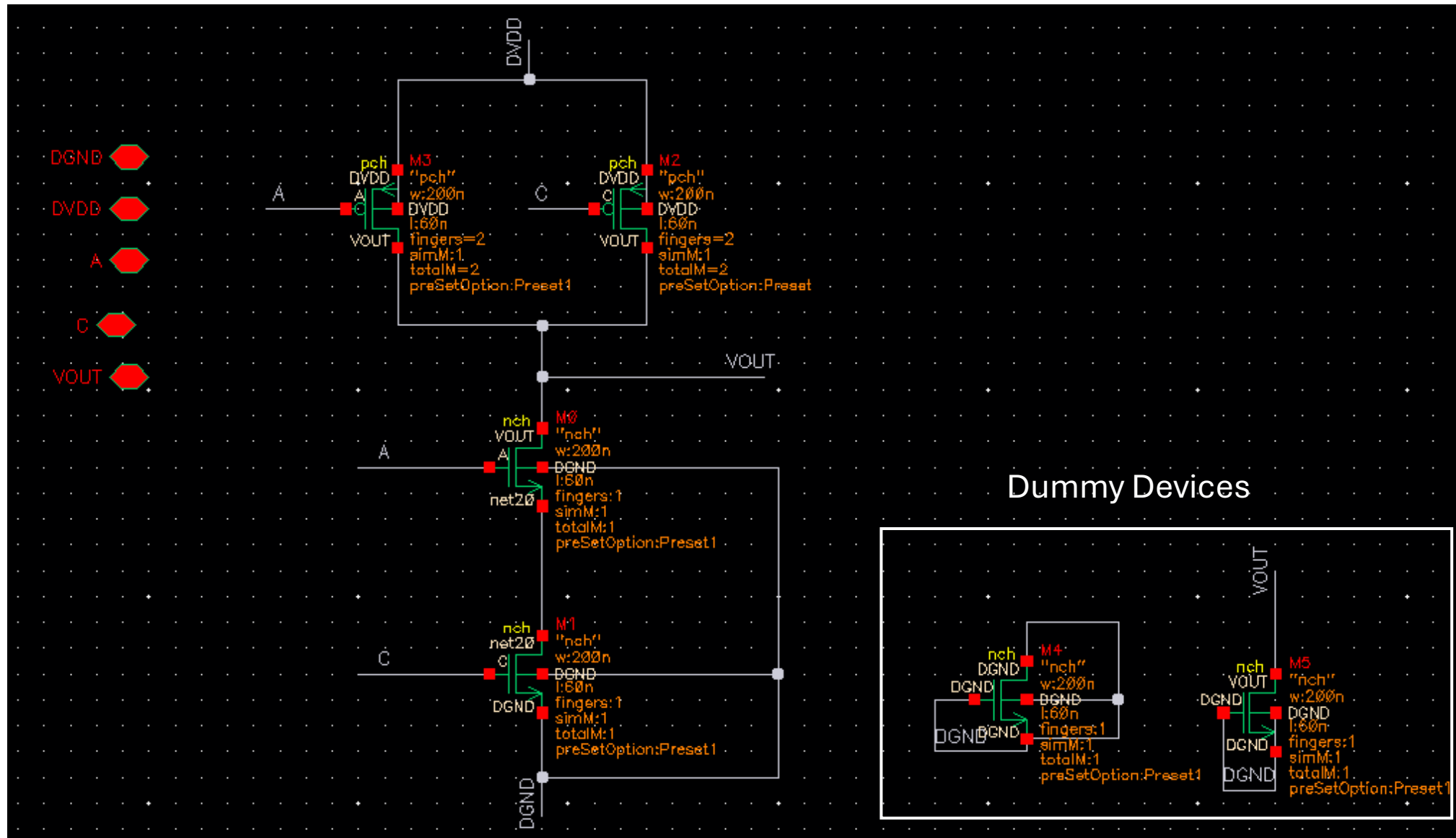


**2-input NAND Gate
Layout
using
TSMC 65nm Technology**

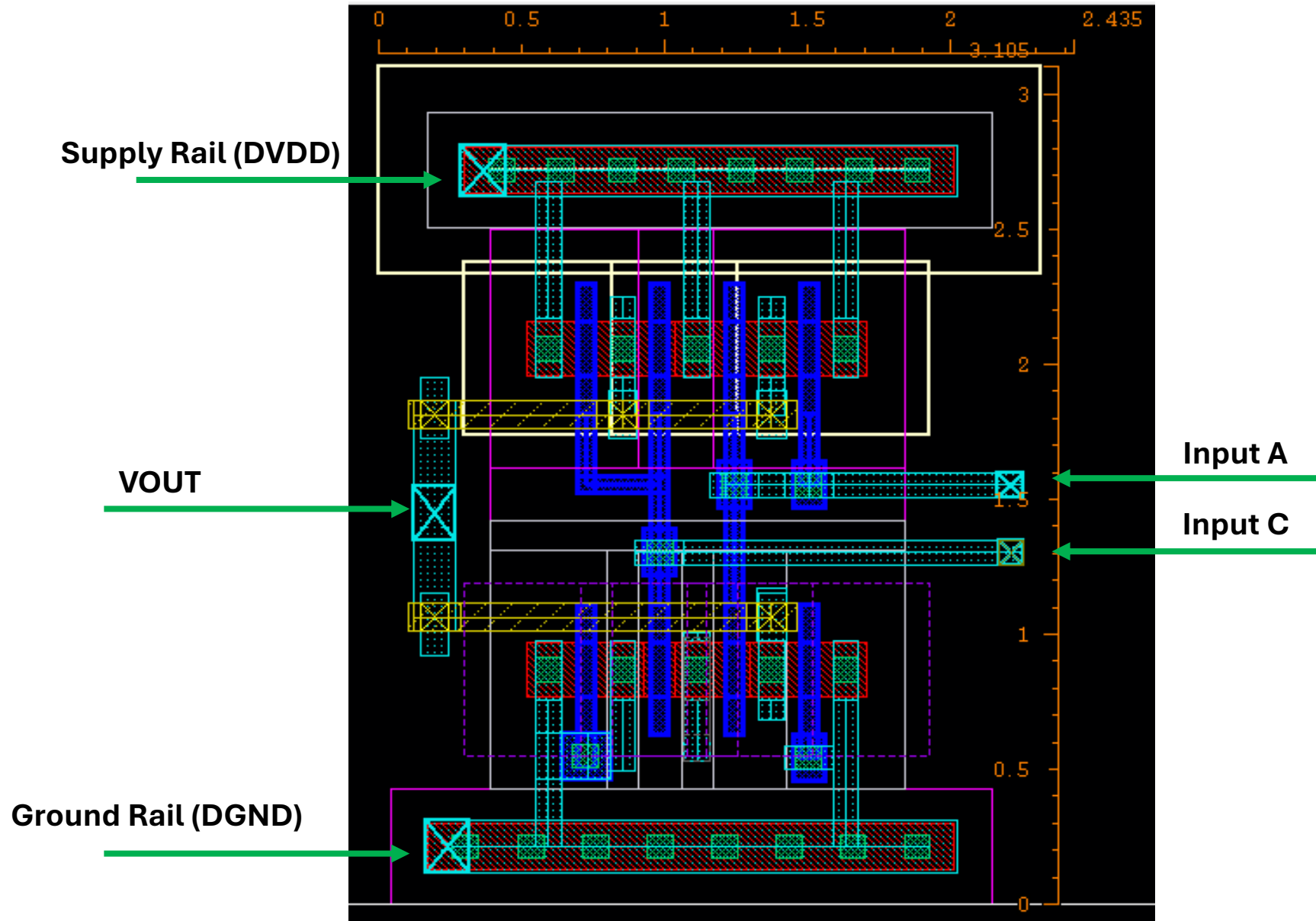
NAND Schematic



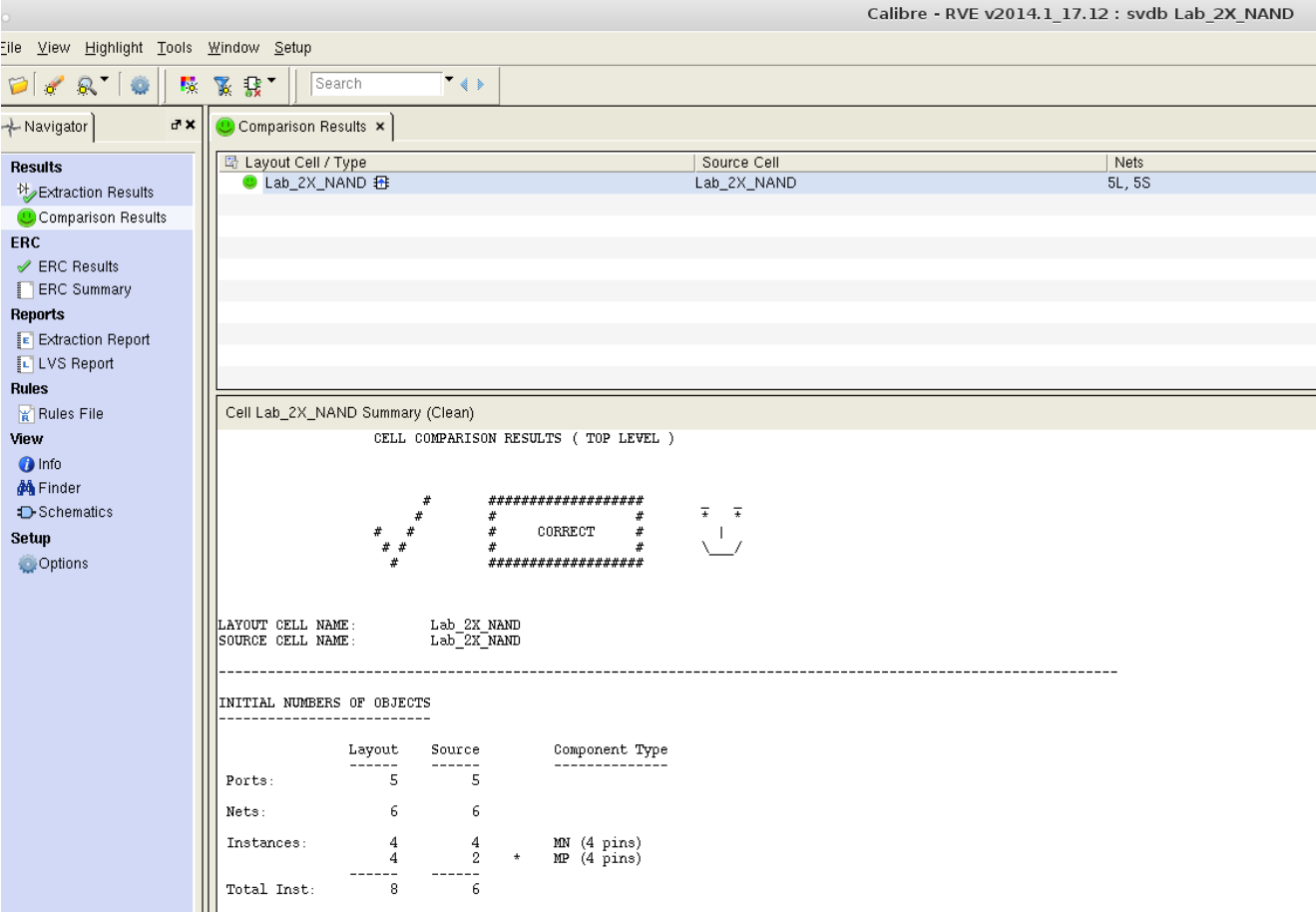
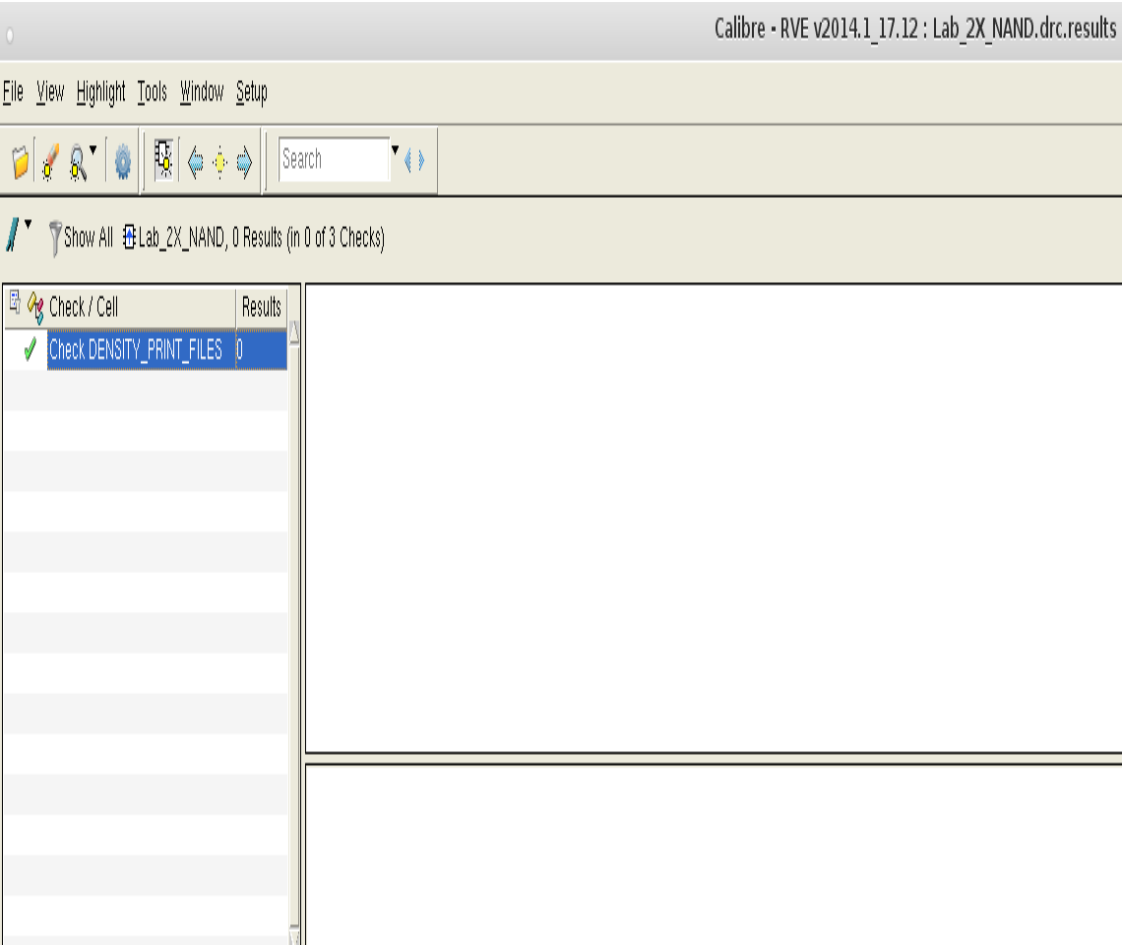
NAND Schematic Notes

- The Two PMOS devices have the same finger width as of NMOS .
- But since the current drive capability of NMOS is higher than that of PMOS because of mobility →
the width of PMOS should be around the double that of the NMOS
- We use number of fingers to double the width of the PMOS devices
- We add two dummies with just one finger to make the number of NMOS devices = Number of PMOS
devices which is 4 transistors.

NAND Layout



DRC and LVS



PEX and Result Viewing Environment (RVE)

Calibre Interactive - PEX v2014.1_17.12

File Transcript Setup

Rules
Inputs
Outputs
PEX Options
Run Control
Transcript
Run PEX
Start RVE

```
--- NETWORK REDUCTION BEGIN:
--- READING FROM PDB...
--- BEGIN REDUCING NETS...
--- DONE REDUCING NETS...
--- WRITING TO PDB...
--- NETWORK REDUCTION COMPLETE: CPU TIME = 0 REAL TIME = 0 LVHEAP = 329/331/331 MALLOC = 340/340/340

-----
PDB NET SUMMARY
-----
pdb file name =          svdb/LAB_2X_NAND.pdb
root cell name =          LAB_2X_NAND
total nets =              6
top-level nets =          6
non-top-level nets =      0
degenerate nets =         0
merged nets =             0
error nets =              0

-----
CALIBRE xRC WARNING / ERROR Summary
-----
xRC Warnings = 1
xRC Errors = 0
-----

--- CALIBRE xRC::FORMATTER COMPLETED - Fri Jul 24 21:11:18 2026
--- TOTAL CPU TIME = 0 REAL TIME = 0 LVHEAP = 26/55/331 MALLOC = 307/307/340 ELAPSED
```

Calibre Info

Calibre View generation completed with 0 WARNINGS and 0 ERRORS.
Please consult the CIW transcript for messages.

Close

9 Warnings

Layer 6 contains unmapped objects and is the source layer of LAYER MAP == 6 DATATYPE == 1.
Layer 6 contains unmapped objects and is the source layer of LAYER MAP == 6 DATATYPE == 3.
Layer 17 contains unmapped objects and is the source layer of LAYER MAP == 17 DATATYPE == 1.
Layer 17 contains unmapped objects and is the source layer of LAYER MAP == 17 DATATYPE == 51.
Layer 31 contains unmapped objects and is the source layer of LAYER MAP == 31 DATATYPE == 1.
Layer 32 contains unmapped objects and is the source layer of LAYER MAP == 32 DATATYPE == 1.
Contact layer nplug cannot appear as one of the connected layers on CONNECT statement. CONNECT statement ignored.
Contact layer pplug cannot appear as one of the connected layers on CONNECT statement. CONNECT statement ignored.
PEX IDEAL XCELL is only applied in gate-level extraction.

Calibre - RVE v2014.1_17.12 : svdb Lab_2X_NAND

File View Highlight Tools Window Setup Help

Search

Navigator

Lab_2X_NAND x

Results

- Extraction Results
- Comparison Results
- Parasitics

Reports

- Extraction Report
- LVS Report

Rules

- Rules File

View

- Info
- Finder
- Schematics

Setup

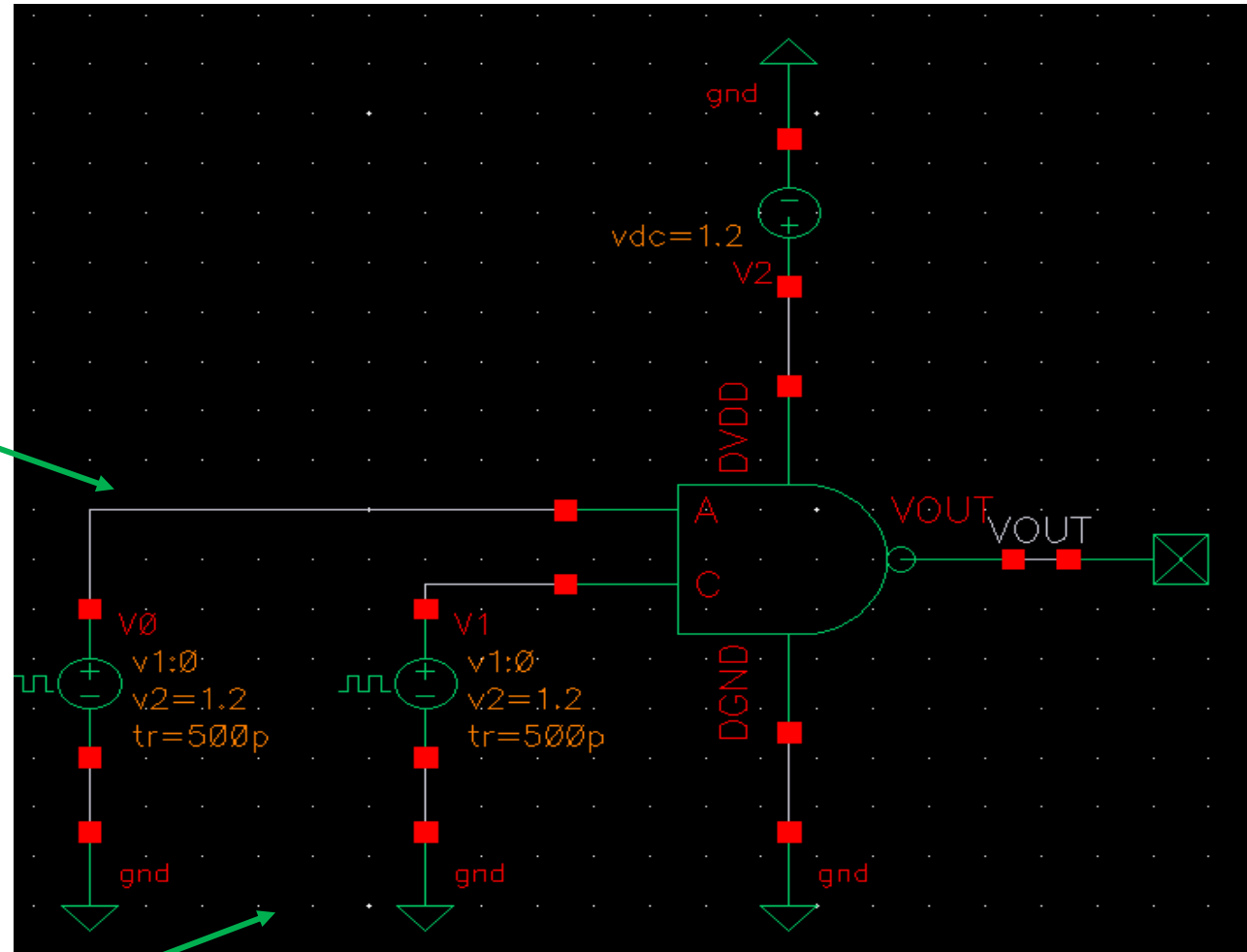
- Options

No.	Layout Net	Source Net	R Count	C Total (F)	CC Total (F)	C+CC Total (F)
1	YOUT	YOUT	30	1.35272E-16	3.52949E-16	4.88220E-16
2	C	C	15	1.08303E-16	3.67770E-16	4.76073E-16
3	3	NET20	2	6.66178E-18	1.20351E-16	1.27012E-16
4	A	A	13	1.23872E-16	3.46782E-16	4.70653E-16
5	DGND	DGND	41	2.01428E-16	1.68957E-16	3.70384E-16
6	DVDD	DVDD	27	1.67278E-16	2.15741E-16	3.83018E-16

NAND Test Bench and Input Setup

DC voltage	1.2 V
AC magnitude	
AC phase	
XF magnitude	
PAC magnitude	
PAC phase	
Voltage 1	0 V
Voltage 2	1.2 V
Period	20n s
Delay time	
Rise time	500p s
Fall time	500p s
Pulse width	10n s

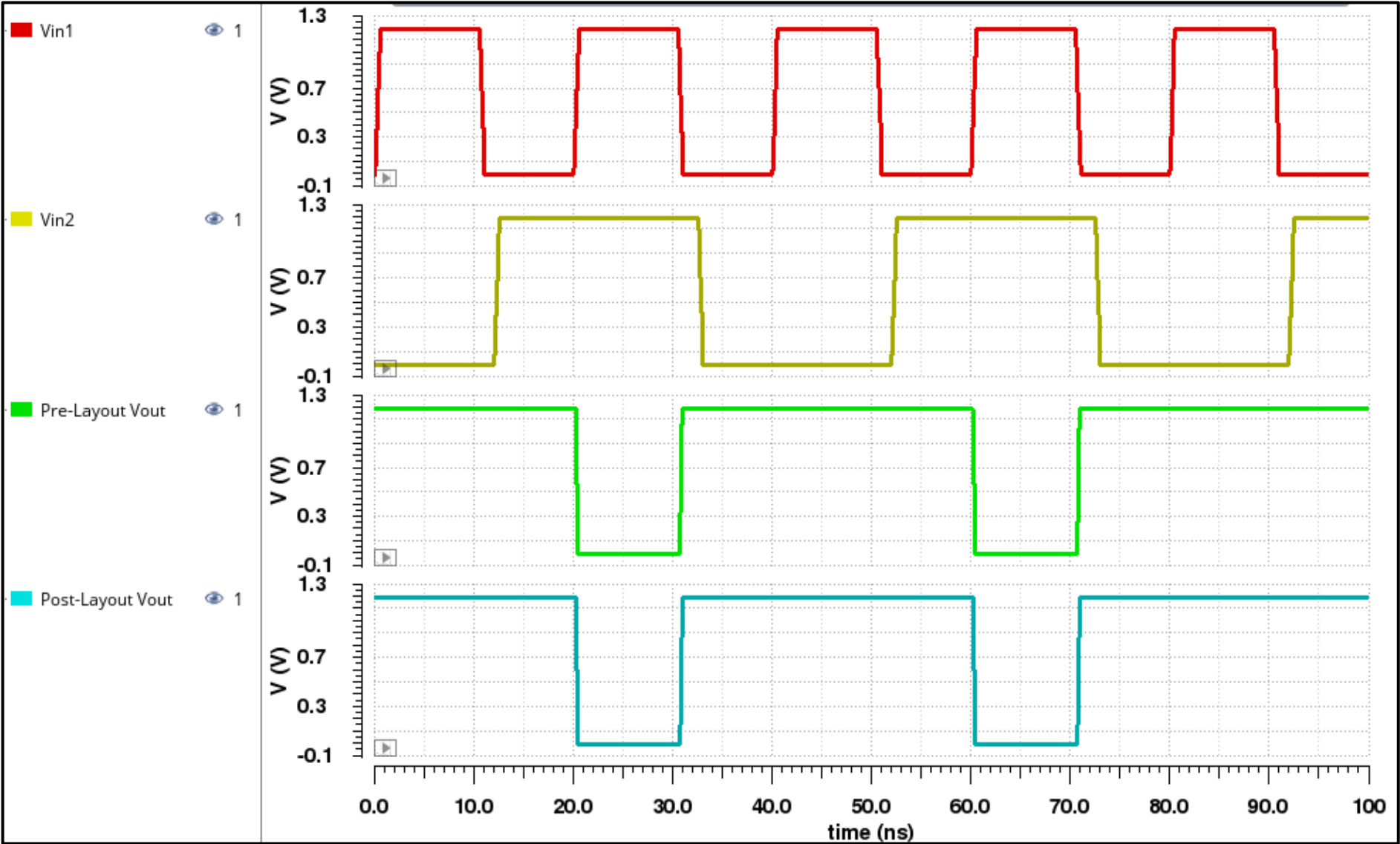
DC voltage	1.2 V
AC magnitude	
AC phase	
XF magnitude	
PAC magnitude	
PAC phase	
Voltage 1	0 V
Voltage 2	1.2 V
Period	40n s
Delay time	12n s
Rise time	500p s
Fall time	500p s
Pulse width	20n s



Simulation Pre-Layout and Post-Layout results with Rise and Fall times

riseTime(VT("/VOUT") 0 nil 1....	74.2p
fallTime(VT("/VOUT") 1.2 nil 0...	79.94p

riseTime(VT("/VOUT") 0 nil 1....	86.38p
fallTime(VT("/VOUT") 1.2 nil 0...	93.85p

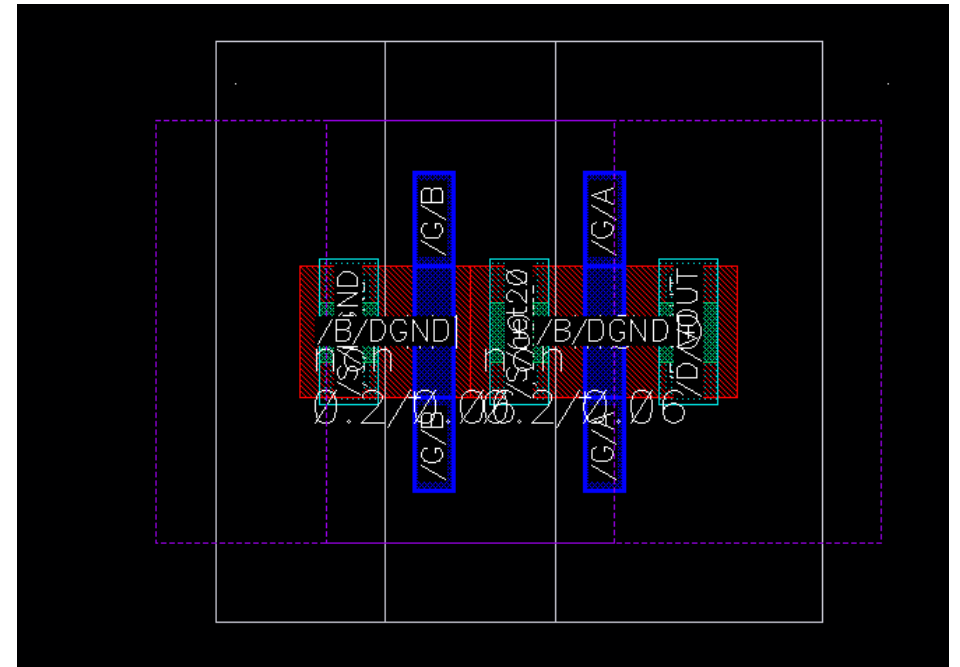
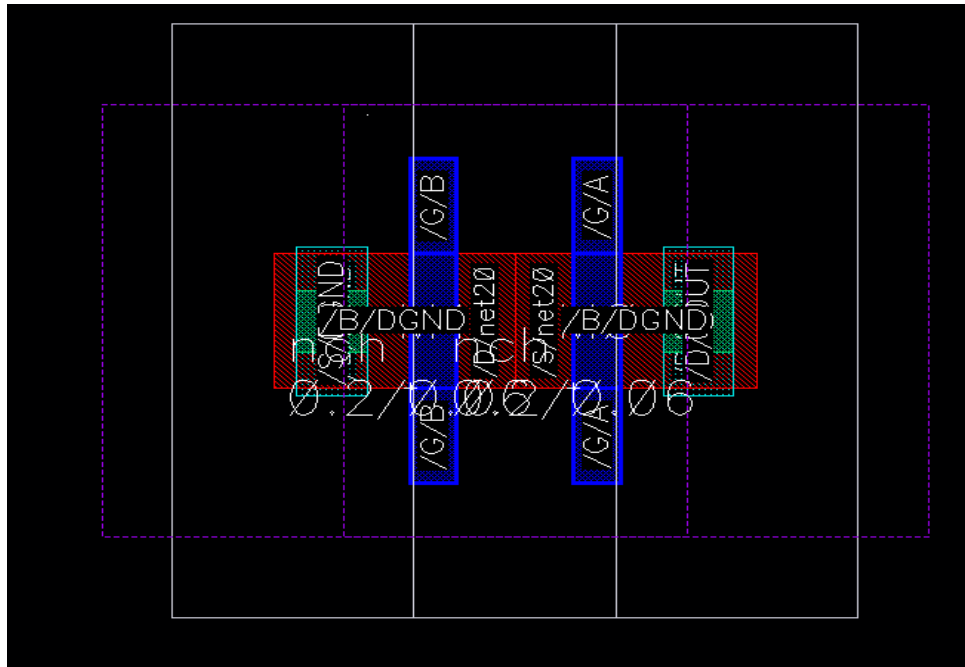


Layout Notes

NAND Layout Notes

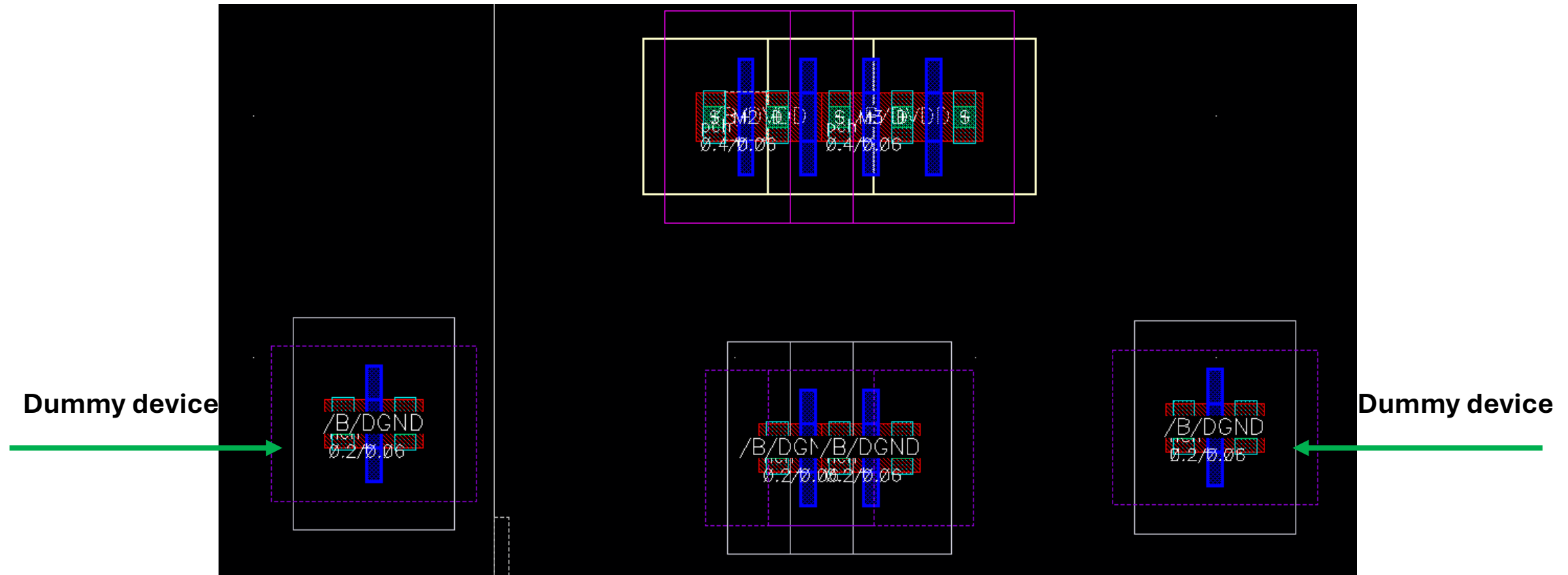
- Latch-up (LUP) is avoided by the well
- When sharing for NMOS, the contact in the middle net disappears this will cause a problem of misalignment with the PMOS devices

To get the contacts select the whole NMOS (the two devices) then right click and add the contact



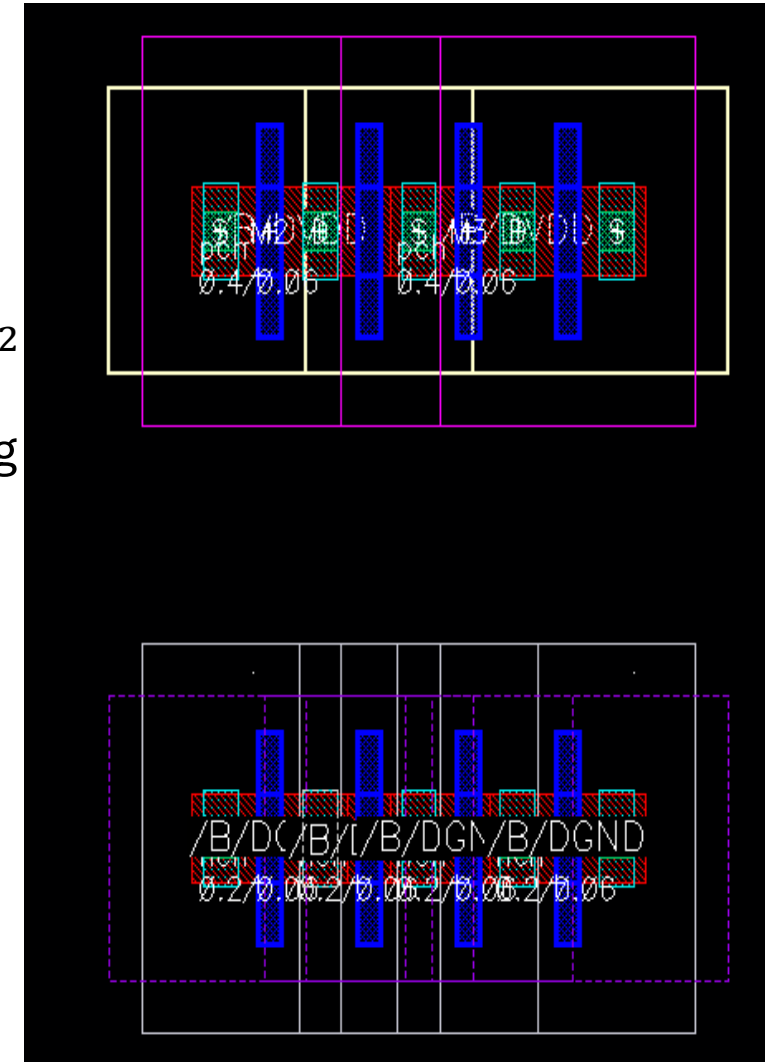
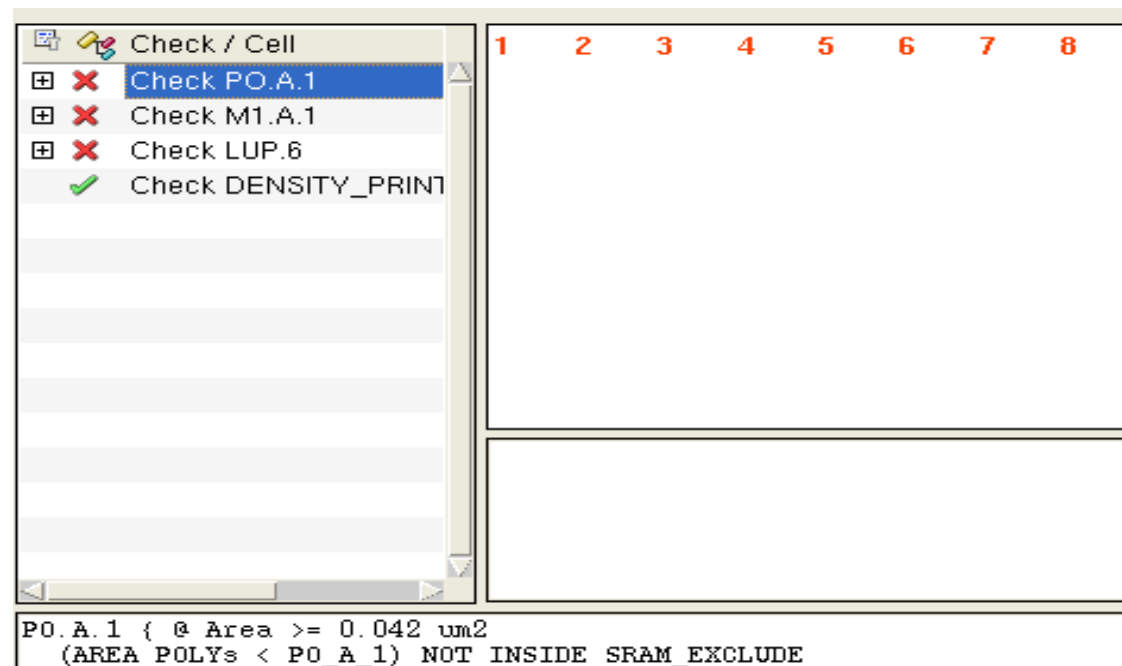
NAND Layout Notes

- The PMOS devices are abutted to each other
- Now the NMOS devices with the dummy ones need to be abutted as well to make align connections between the two transistors



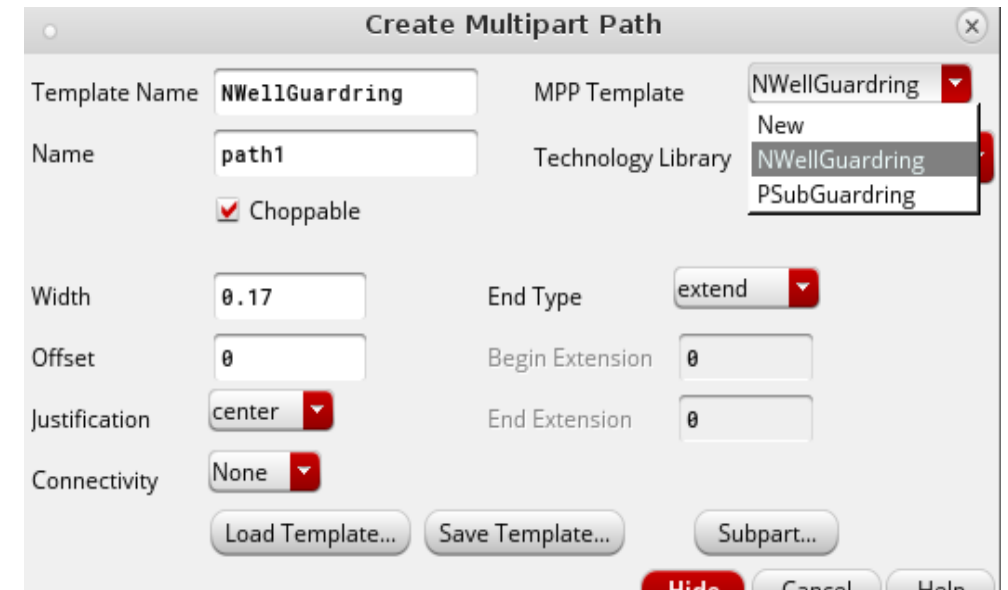
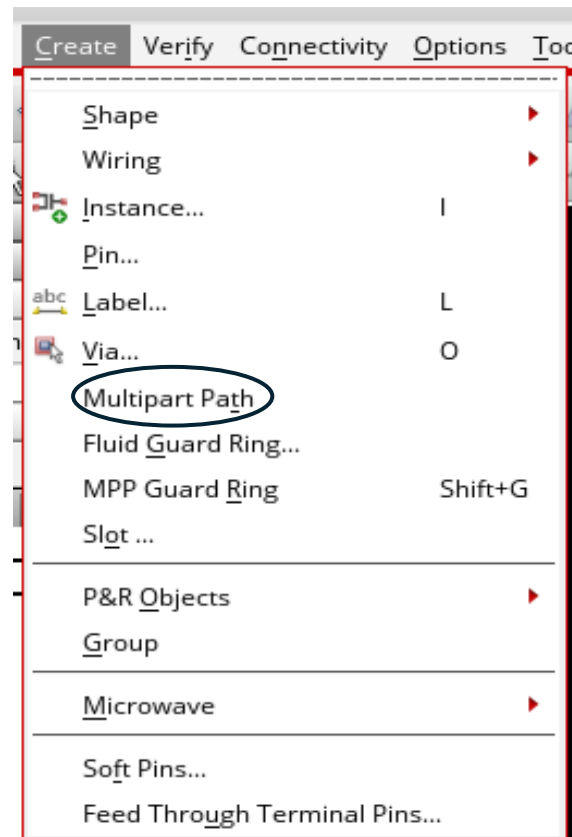
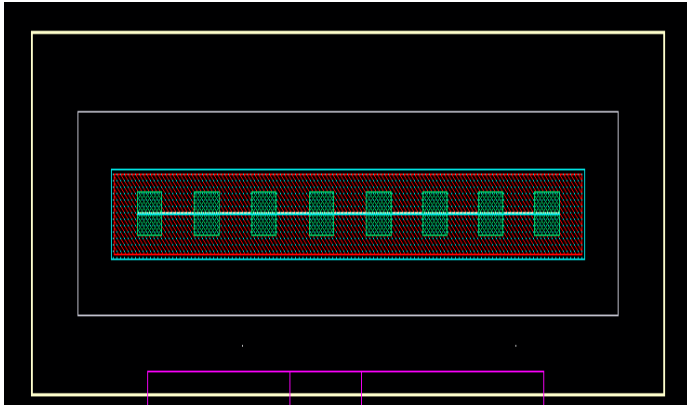
NAND Layout Notes

- After sharing the NMOS with dummies and PMOS , DRC is run
- There will be 3 errors
- One error for the LUP effect → solved by Nwell
- Two errors are for the area → PO Area and M1 Area should be $> 0.042 \mu m^2$
- These area errors solved by increasing area or can be solved while routing



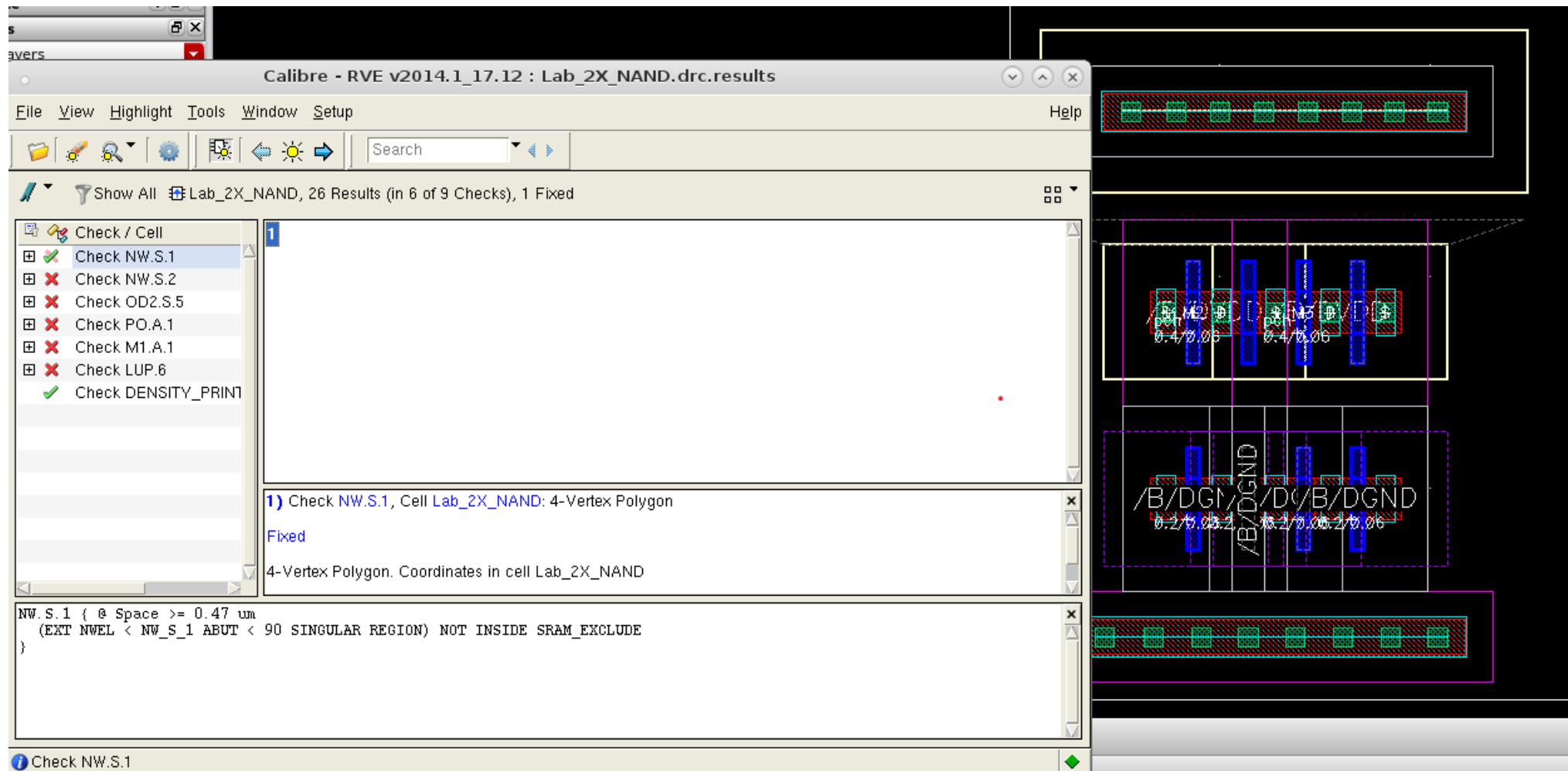
NAND Layout Notes

- To align to devices or components press A on the two sides to be aligned
- To create Vias → press p on the metal layer then press space → select the type of via you want with routing needed
- To create the VDD or GND rail → from create in the tool bar → multipart path → press f3 → from MPP template → for VDD ,select NWell guarding and for GND select PSub guarding



NAND Layout Notes

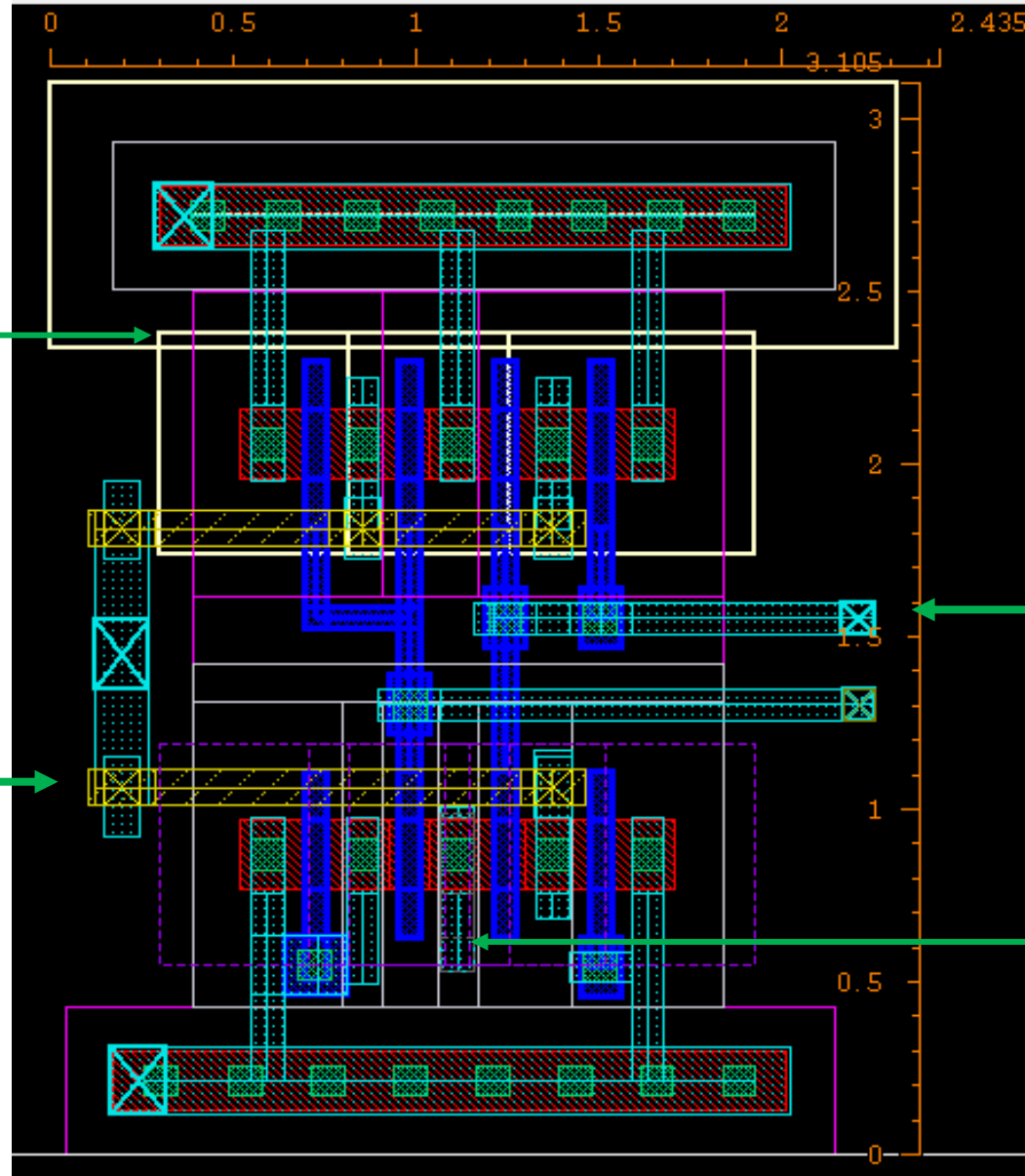
This DRC means the two Nwells should overlap on each other



NAND Layout Notes

The NWELL of DVDD must overlap with NWELL of PMOS

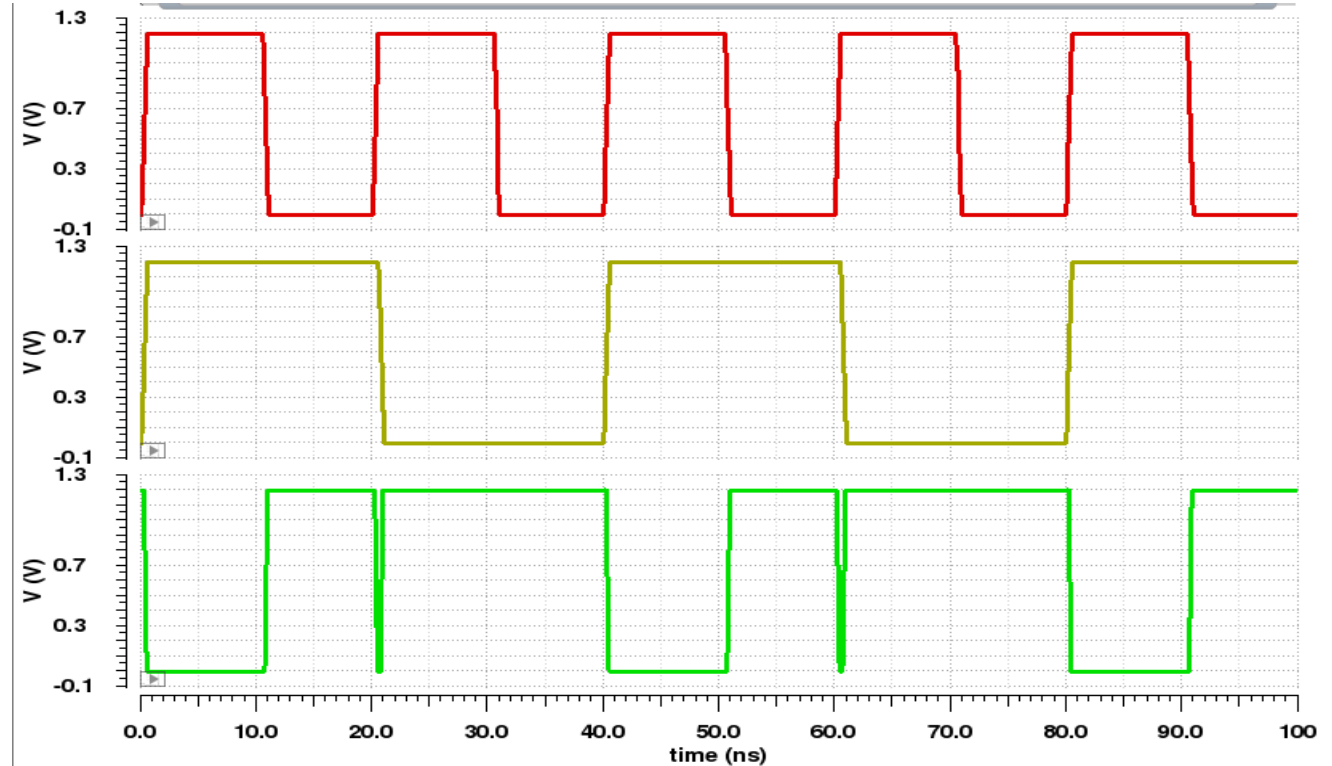
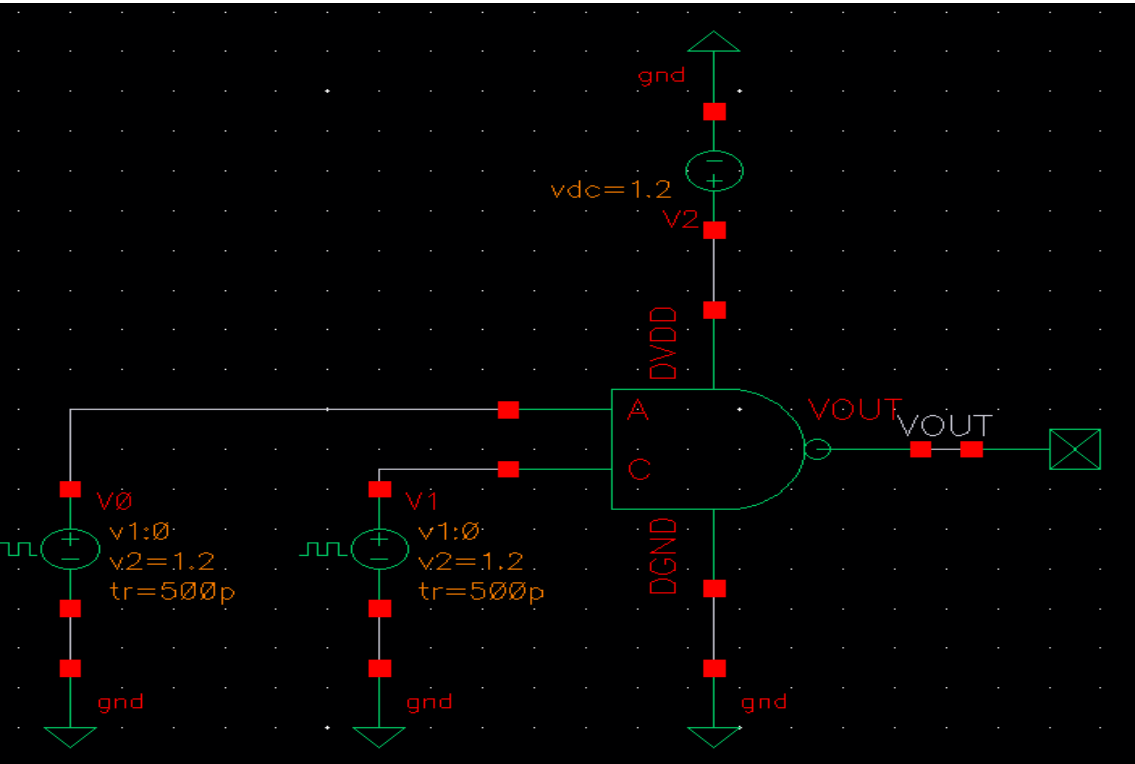
M2 was used to try to achieve Manhattan routing technique



It would be better if the horizontal routing was in M2 layer. The problem I made was that I didn't increase the distance between the devices PMOs and NMOS → Vias of M2 made problem of spacing (should be considered)

- This is the shared node between the two NMOS devices (net20)
- M1 of this node is increased to pass the DRC which is Area of M1 $> 0.042 \mu m^2$

NAND Layout Notes



- To avoid race condition when input1 goes from low to high and input2 from high to low → put delay in input2 of 12 ps